Technology Stack

Date	07 July 2026
Project Name	CaptionAI: AI Powered Social Media Caption Generator
Maximum Marks	2 Marks

Define Your Technology Stack:

This section identifies the programming languages, frameworks, databases, front-end tools, back-end tools, and APIs used to build CaptionAI. The chosen technology stack ensures that the application is scalable, maintainable, and aligned with the project requirements.

Step 1: Technology Stack

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	Frontend / Client-Side	HTML5, CSS3, JavaScript	HTML5 provides the structure of the web pages, CSS3 creates an attractive and responsive user interface, and JavaScript adds interactivity such as form validation, dynamic updates, and user-friendly features.
2	Backend / Server-Side	Python (Flask)	Flask is a lightweight Python web framework used to handle user requests, connect with the AI model, manage application logic, and serve web pages efficiently.
3	Database / Data Storage	SQLite	SQLite stores user account information and caption history. It is easy to configure, lightweight, and suitable for small to medium-sized web applications.
4	Cloud / Deployment	Ngrok (Development) and Render (Deployment)	Ngrok was used to expose the local Flask server during development and testing. Render is used for deploying the application online so that users can access it through a web browser.
5	Version Control	Git & GitHub	Git tracks changes made to the source code, while GitHub provides version control, collaboration among team members, and secure project backup.
6	AI Model / External Service	Groq API (LLaMA 3.3-70B Versatile)	The Groq API processes user prompts and generates high-quality captions, relevant hashtags, and call-to-action suggestions with fast response times.

Step 2: Technology Architecture Summary

The CaptionAI application follows a client-server architecture. Users interact with the frontend developed using HTML, CSS, and JavaScript. The frontend sends requests to the Flask backend, which communicates with the Groq API to generate AI-powered captions. User details and caption history are stored in the SQLite database. GitHub is used for version control, while Ngrok and Render support testing and deployment.

Technologies Used (Quick Summary)

Frontend	HTML5, CSS3, JavaScript
Backend	Python (Flask)
Database	SQLite
AI Model	Groq API (LLaMA 3.3-70B Versatile)
Version Control	Git & GitHub
Development Tool	Visual Studio Code
Deployment	Ngrok (Testing), Render (Production)